Balanced Diet & Biotechnology – Page 424

Source: Lucent's General Knowledge – Biology

Biology: Diet, Biotechnology & Applied Biology

50 Questions • 30 Minutes

Instructions:
• Each question has 4 options (A, B, C, D).
• The correct answer is shown on the right side of each question.
• Detailed explanations with textbook page references are at the end.

Questions

Q1. Necessary calorie requirement for a male light worker is: **A.** 3000 calorie **B.** 2000 calorie **C.** 3600 calorie **D.** 2500 calorie	**Ans: B**
Q2. Necessary calorie requirement for a female eight-hour worker is: **A.** 2000 calorie **B.** 3000 calorie **C.** 2500 calorie **D.** 3600 calorie	**Ans: C**
Q3. A male hard worker requires how many calories? **A.** 2000 **B.** 2500 **C.** 3000 **D.** 3600	**Ans: D**
Q4. Which substance is used as a preservative to preserve food items? **A.** Sodium chloride **B.** Sodium benzoate **C.** Calcium carbonate **D.** Potassium permanganate	**Ans: B**
Q5. Milk is not considered a complete food due to lack of: **A.** Calcium and Protein **B.** Vitamin C and Iron **C.** Vitamin A and D **D.** Fat and Carbohydrate	**Ans: B**
Q6. Which is a genetically modified vegetable recently made available in Indian market? **A.** Bt-Cotton **B.** Bt-Brinjal **C.** Bt-Corn **D.** Bt-Potato	**Ans: B**
Q7. Biotechnology deals with techniques of using: **A.** Only chemicals **B.** Live organisms or enzymes to produce useful products **C.** Only metals **D.** Nuclear energy	**Ans: B**

Q8. Genetic Engineering is the technique to alter: **A.** Physical structure of organism **B.** Chemistry of genetic material (DNA & RNA) to change phenotype **C.** Only protein structure **D.** Only cell membrane	**Ans: B**

Q9. Bioprocess Engineering involves maintenance of: **A.** Acidic environment **B.** Sterile ambience for desired microbes/eukaryotic cells **C.** High temperature only **D.** Vacuum conditions	**Ans: B**

Q10. Restriction enzyme is known as: **A.** Molecular glue **B.** Molecular scissors **C.** Molecular hammer **D.** Molecular tape	**Ans: B**

Q11. Plasmid is: **A.** A type of virus **B.** Extra small circular DNA in bacteria **C.** A type of protein **D.** Part of ribosome	**Ans: B**

Q12. Due to presence of plasmid, bacteria show resistance to: **A.** Heat **B.** Many antibiotics **C.** UV radiation **D.** Cold temperature	**Ans: B**

Q13. A plasmid can be used as a vector to deliver: **A.** Protein into host cell **B.** An alien piece of DNA into host organism **C.** Carbohydrate **D.** Lipids	**Ans: B**

Q14. DNA ligase is an enzyme that: **A.** Cuts DNA at specific sites **B.** Joins cut ends of DNA molecule together **C.** Unwinds DNA **D.** Copies DNA	**Ans: B**

Q15. Recombinant DNA is created by: **A.** Destroying DNA **B.** Combining two fragments of DNA from two different sources **C.** Heating DNA **D.** Freezing DNA	**Ans: B**

Q16. Cloning is the process of producing: **A.** Different versions of DNA **B.** Large number of identical copies of any template DNA **C.** Mutated DNA **D.** Shortened DNA	**Ans: B**
Q17. Restriction enzymes belong to a larger class of enzymes called: **A.** Ligases **B.** Nucleases **C.** Proteases **D.** Lipases	**Ans: B**
Q18. Exonuclease removes nucleotide from: **A.** The middle of DNA **B.** The end of DNA **C.** Both ends simultaneously **D.** Only RNA	**Ans: B**
Q19. Endonuclease cuts at: **A.** End of DNA only **B.** Specific position within the DNA **C.** Random positions **D.** Only RNA	**Ans: B**
Q20. Organisms whose genes have been altered by manipulation are called: **A.** Hybrid organisms **B.** Genetically Modified Organisms (GMO) **C.** Cloned organisms **D.** Mutant organisms	**Ans: B**
Q21. Gene therapy is: **A.** Treatment using antibiotics **B.** Collection of methods that allow correction of a gene defect in a child or embryo **C.** Surgical operation on genes **D.** Radiation therapy	**Ans: B**
Q22. The first clinical gene therapy was given in 1990 to a four-year-old girl with which deficiency? **A.** Insulin deficiency **B.** Adenosine deaminase (ADA) deficiency **C.** Iron deficiency **D.** Vitamin D deficiency	**Ans: B**

Q23. Genetically modified plants are useful in: **A.** Decreasing crop yields **B.** Increasing crop yields, reducing losses, and making crops stress tolerant **C.** Increasing pesticide use **D.** Reducing soil fertility	**Ans: B**
Q24. Papaver somniferum plant yields: **A.** Antibiotics **B.** Powerful analgesic **C.** Vitamins **D.** Minerals	**Ans: B**
Q25. The major source of sugar in the world is: **A.** Cocos nucifera **B.** Saccharum Officinarum **C.** Papaver somniferum **D.** Bt-Brinjal	**Ans: B**
Q26. Cocos nucifera is widely used in: **A.** Making medicines **B.** Preparation of culture medium **C.** Making clothes **D.** Building materials	**Ans: B**
Q27. To yield milk, cow is given which hormone? **A.** Insulin **B.** Prolactin **C.** Thyroxin **D.** Adrenaline	**Ans: B**
Q28. The best way to determine paternity is: **A.** Blood group test **B.** DNA finger printing **C.** X-ray **D.** CT scan	**Ans: B**
Q29. Chloromycetin is obtained from: **A.** Penicillium **B.** Streptomycetin venezuelae **C.** E. coli **D.** Yeast	**Ans: B**
Q30. In Recombinant Vector Vaccines, which are used as vectors? **A.** Chemicals **B.** Bacteria and virus **C.** Proteins **D.** Minerals	**Ans: B**

Q31. Bollgard I and Bollgard II technology are used in: **A.** Making vaccines **B.** Developing genetically modified crop plants **C.** Water purification **D.** Soil testing	**Ans: B**
Q32. The term 'ACE-2' is talked about in the context of: **A.** Cancer treatment **B.** Spread of viral diseases **C.** Diabetes management **D.** Heart surgery	**Ans: B**
Q33. A diagnostic test, Real Time PCR for COVID-19 detects: **A.** Protein of virus **B.** RNA of virus **C.** DNA of bacteria **D.** Antibodies only	**Ans: B**
Q34. Rapid Antibodies Test does not require sample from nose and throat and was approved by: **A.** WHO **B.** Indian Council of Medical Research **C.** FDA **D.** CDC	**Ans: B**
Q35. Vector vaccines work by: **A.** Killing all viruses directly **B.** Entering cells and enabling them to create spike protein **C.** Blocking all receptors **D.** Destroying antibodies	**Ans: B**
Q36. Sinovac given for COVID-19 is a: **A.** mRNA vaccine **B.** Whole virus vaccine **C.** Vector vaccine **D.** DNA vaccine	**Ans: B**
Q37. DNA Barcoding is a tool to: **A.** Create new DNA **B.** Distinguish among species that look alike **C.** Destroy harmful DNA **D.** Clone organisms	**Ans: B**

Q38. Covishield vaccine was produced by Serum Institute of India using which platform?

- **A.** Whole virus platform
- **B.** mRNA platform
- **C.** DNA platform
- **D.** Protein subunit platform

Ans: B

Q39. Sputnik V vaccine is manufactured by using:

- **A.** mRNA platform
- **B.** Vector base platform
- **C.** Whole virus
- **D.** Protein subunit

Ans: B

Q40. Applications of biotechnology include:

- **A.** Only medicine
- **B.** Agriculture, therapeutic, diagnostic, biopharmaceutical, processed food, waste treatment, energy production
- **C.** Only agriculture
- **D.** Only energy

Ans: B

Q41. Calorie requirement for a female light worker is:

- **A.** 2000
- **B.** 2100
- **C.** 2500
- **D.** 3000

Ans: B

Q42. Calorie requirement for a male eight-hour worker is:

- **A.** 2000
- **B.** 2500
- **C.** 3000
- **D.** 3600

Ans: C

Q43. Calorie requirement for a female hard worker is:

- **A.** 2100
- **B.** 2500
- **C.** 3000
- **D.** 3600

Ans: C

Q44. In the balanced diet chart, the grain (wheat, rice) requirement for an adult male doing normal work is:

- **A.** 400 g
- **B.** 520 g
- **C.** 670 g
- **D.** 410 g

Ans: B

Q45. Making curd, bread or wine are examples of: **A.** Genetic engineering **B.** Biotechnology **C.** Gene therapy **D.** Cloning	**Ans: B**
Q46. Which technology creates unique character (antibiotic resistance) in bacteria? **A.** Gene therapy **B.** Plasmid presence **C.** Cloning **D.** PCR	**Ans: B**
Q47. Restriction enzyme cuts DNA at: **A.** Random locations **B.** Specific location **C.** Only at the ends **D.** Only the middle	**Ans: B**
Q48. The milk requirement for an adult male doing hard work is: **A.** 150 g **B.** 200 g **C.** 250 g **D.** 100 g	**Ans: C**
Q49. Pulses requirement for adult male normal worker is: **A.** 40 g **B.** 50 g **C.** 60 g **D.** 80 g	**Ans: B**
Q50. An example of GMO food crop approved in India is: **A.** Bt-Rice **B.** Bt-Brinjal **C.** Bt-Wheat **D.** Bt-Mango	**Ans: B**

Answer Key & Explanations

Q1. Answer: B — 2000 calorie

Page 424: Light worker male requires 2000 calorie, female 2100 calorie.

Topic: Balanced Diet

Q2. Answer: C — 2500 calorie

Page 424: Eight hours worker female requires 2500 calorie, male 3000 calorie.

Topic: Balanced Diet

Q3. Answer: D — 3600

Page 424: Hard worker male requires 3600 calorie, female 3000 calorie.

Topic: Balanced Diet

Q4. Answer: B — Sodium benzoate

Page 424: Sodium Benzoate is used as preservative to preserve food items.

Topic: Food Science

Q5. Answer: B — Vitamin C and Iron

Page 424: Milk is not considered as complete food due to lack of vitamin C and iron.

Topic: Food Science

Q6. Answer: B — Bt-Brinjal

Page 424: Bt-Brinjal is a genetically modified vegetable recently being made available in Indian market.

Topic: Biotechnology

Q7. Answer: B — Live organisms or enzymes to produce useful products

Page 424: Biotechnology deals with techniques of using live organisms or enzymes from organisms to produce useful products (e.g., curd, bread, wine).

Topic: Biotechnology

Q8. Answer: B — Chemistry of genetic material (DNA & RNA) to change phenotype

Page 424: Genetic Engineering is the technique to alter the chemistry of genetic material (DNA & RNA) and introduce into host organism, changing the phenotype.

Topic: Biotechnology

Q9. Answer: B — Sterile ambience for desired microbes/eukaryotic cells

Page 424: Bioprocess Engineering is maintenance of sterile ambience for growth of desired microbes or eukaryotic cells for manufacture of biotechnological products.

Topic: Biotechnology

Q10. Answer: B — Molecular scissors

Page 424: Restriction enzyme is known as molecular scissors cutting the DNA at specific location.

Topic: Biotechnology

Q11. Answer: B — Extra small circular DNA in bacteria

Page 424: In addition to genomic DNA, many bacteria have extra small circular DNA known as plasmid.

Topic: Biotechnology

Q12. Answer: B — Many antibiotics

Page 424: Due to presence of plasmid, bacteria show resistance to many antibiotics. This creates a unique character in bacteria.

Topic: Biotechnology

Q13. Answer: B — An alien piece of DNA into host organism

Page 424: A plasmid can be used as vector to deliver an alien piece of DNA into host organism.

Topic: Biotechnology

Q14. Answer: B — Joins cut ends of DNA molecule together

Page 424: DNA ligase is an enzyme by which cut ends of DNA molecule join together.

Topic: Biotechnology

Q15. Answer: B — Combining two fragments of DNA from two different sources

Page 424: Recombinant DNA is a DNA which has been created by combining two fragments of DNA from two different sources.

Topic: Biotechnology

Q16. Answer: B — Large number of identical copies of any template DNA

Page 424: Cloning is producing large number of identical copies of any template DNA.

Topic: Biotechnology

Q17. Answer: B — Nucleases

Page 424: Restriction enzymes belong to larger class of enzymes called nucleases.

Topic: Biotechnology

Q18. Answer: B — The end of DNA

Page 424: Exonuclease removes nucleotide from the end of DNA.

Topic: Biotechnology

Q19. Answer: B — Specific position within the DNA

Page 424: Endonuclease cuts at specific position within the DNA.

Topic: Biotechnology

Q20. Answer: B — Genetically Modified Organisms (GMO)

Page 424: Plant, bacteria, fungi and animals whose genes have been altered by manipulation are called Genetically Modified Organisms (GMO).

Topic: Biotechnology

Q21. Answer: B — Collection of methods that allow correction of a gene defect in a child or embryo

Page 424: Gene therapy is a collection of methods that allow correction of a gene defect that has been diagnosed in a child or embryo.

Topic: Biotechnology

Q22. Answer: B — Adenosine deaminase (ADA) deficiency

Page 424: The first clinical gene therapy was given in 1990 to a four-year-old girl with adenosine deaminase (ADA) deficiency.

Topic: Biotechnology

Q23. Answer: B — Increasing crop yields, reducing losses, and making crops stress tolerant

Page 424: GM plants are useful in increasing crop yields, reduce losses and make the crop more tolerant to stresses.

Topic: Biotechnology

Q24. Answer: B — Powerful analgesic

Page 424: Papaver somniferum plant yields powerful analgesic.

Topic: Biotechnology

Q25. Answer: B — Saccharum Officinarum

Page 424: Major source of sugar in the world is Saccharum Officinarum.

Topic: Biotechnology

Q26. Answer: B — Preparation of culture medium

Page 424: Cocos nucifera is a plant material widely used in preparation of culture medium.

Topic: Biotechnology

Q27. Answer: B — Prolactin

Page 424: To yield milk, cow is given prolactin.

Topic: Biotechnology

Q28. Answer: B — DNA finger printing

Page 424: The best way to determine paternity is DNA finger printing.

Topic: Biotechnology

Q29. Answer: B — Streptomycetin venezuelae

Page 424: Chloromycetin is obtained from streptomycetin venezuelae.

Topic: Biotechnology

Q30. Answer: B — Bacteria and virus

Page 424: Genetic engineering is applied in development of Recombinant Vector Vaccines in which bacteria and virus are used as vectors.

Topic: Biotechnology

Q31. Answer: B — Developing genetically modified crop plants

Page 424: Bollgard I and Bollgard II technology are used in developing genetically modified crop plants.

Topic: Biotechnology

Q32. Answer: B — Spread of viral diseases

Page 424: The term ACE-2 is talked about in the context of spread of viral diseases.

Topic: Biotechnology

Q33. Answer: B — RNA of virus

Page 424: Real Time PCR for COVID-19 detects RNA of virus.

Topic: Biotechnology

Q34. Answer: B — Indian Council of Medical Research

Page 424: Rapid Antibodies Test does not require sample from nose and throat in COVID-19 test, approved by Indian Council of Medical Research.

Topic: Biotechnology

Q35. Answer: B — Entering cells and enabling them to create spike protein

Page 424: Vector vaccines work to provide immunity by entering directly into the cells and enabling them to create spike protein.

Topic: Biotechnology

Q36. Answer: B — Whole virus vaccine

Page 424: Sinovac given for Covid-19 is a whole virus vaccine.

Topic: Biotechnology

Q37. Answer: B — Distinguish among species that look alike

Page 424: DNA Barcoding can be a tool to distinguish among species that look alike.

Topic: Biotechnology

Q38. Answer: B — mRNA platform

Page 424: The Serum Institute of India produced Covishield using mRNA platform.

Topic: Biotechnology

Q39. Answer: B — Vector base platform

Page 424: Sputnik V vaccine is manufactured by using vector base platform.

Topic: Biotechnology

Q40. Answer: B — Agriculture, therapeutic, diagnostic, biopharmaceutical, processed food, waste treatment, energy production

Page 424: Applications include agriculture, therapeutic, diagnostic, biopharmaceutical, processed food, waste treatment and energy production.

Topic: Biotechnology

Q41. Answer: B — 2100

Page 424: Female light worker requires 2100 calories.

Topic: Balanced Diet

Q42. Answer: C — 3000

Page 424: Male eight-hour worker requires 3000 calories.

Topic: Balanced Diet

Q43. Answer: C — 3000
Page 424: Female hard worker requires 3000 calories.

Topic: Balanced Diet

Q44. Answer: B — 520 g
Page 424: For adult male moderate (M) worker, grain requirement is 520 g.

Topic: Balanced Diet

Q45. Answer: B — Biotechnology
Page 424: Biotechnology examples include making curd, bread or wine in which microbes are used.

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Q46. Answer: B — Plasmid presence
Page 424: Due to presence of plasmid, bacteria show resistance to many antibiotics, creating unique character.

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Q47. Answer: B — Specific location
Page 424: Restriction enzyme cuts the DNA at specific location.

Topic: Biotechnology

Q48. Answer: C — 250 g
Page 424: For adult male hard worker, milk requirement is 250 g.

Topic: Balanced Diet

Q49. Answer: B — 50 g
Page 424: For adult male moderate worker, pulses requirement is 50 g.

Topic: Balanced Diet

Q50. Answer: B — Bt-Brinjal
Page 424: Bt-Brinjal is a genetically modified vegetable recently made available in Indian market.

Topic: Biotechnology